

MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Final Internship Report

Author

Tumul Singh

Institute

National Institute of Technology Kurukshetra

Project Duration

Bluestock Data Analytics Internship

1. Executive Summary

The Mutual Fund Analytics Capstone Project was developed to analyze multiple aspects of the Indian mutual fund industry using data analytics, SQL, Python, and Power BI. The project integrates data from multiple sources including fund master data, NAV history, AUM trends, SIP inflows, category inflows, investor transactions, portfolio holdings, and benchmark indices.

The primary objective was to build an end-to-end analytics pipeline capable of transforming raw financial data into actionable business insights. The project includes data cleaning, exploratory data analysis, performance evaluation, portfolio analysis, investor behavior analysis, and dashboard development.

The final output consists of an interactive Power BI dashboard with five analytical modules, a structured analytical repository, and a comprehensive reporting framework.

2. Project Objectives

The key objectives of this project were:

- Analyze mutual fund industry growth and trends.
 - Evaluate fund performance using risk-adjusted metrics.
 - Study investor transaction behavior and demographics.
 - Analyze portfolio holdings and sector allocations.
 - Compare benchmark index performance.
 - Build a professional Power BI dashboard.
 - Generate actionable insights for investment decision-making.
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3. Dataset Description

The project utilizes ten cleaned datasets.

3.1 Fund Master Dataset

Contains scheme-level information including:

- AMFI Code
- Scheme Name
- Fund House
- Category
- Plan Type

3.2 NAV History Dataset

Contains historical NAV records used for trend analysis and return calculations.

3.3 AUM Dataset

Contains Assets Under Management data for different fund houses across multiple periods.

3.4 SIP Inflow Dataset

Tracks monthly SIP inflows and SIP account growth.

3.5 Category Inflow Dataset

Provides category-wise net inflows into mutual fund schemes.

3.6 Industry Folio Dataset

Contains folio counts across equity, debt, hybrid, and other categories.

3.7 Scheme Performance Dataset

Contains:

- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio
- Returns
- Risk Grades

3.8 Investor Transactions Dataset

Contains:

- Investor IDs
- Gender
- Age Group
- Income Information

- Transaction Details

3.9 Portfolio Holdings Dataset

Contains:

- Stock Name
- Sector
- Weight Percentage
- Market Value

3.10 Benchmark Dataset

Contains benchmark index values and historical performance data.

4. ETL Pipeline Design

The project follows a structured ETL architecture.

Extract

Raw datasets were collected from multiple financial data sources.

Transform

Data transformation activities included:

- Missing value handling
- Column standardization
- Data type conversion
- Duplicate removal
- Date formatting
- Feature engineering

Load

Cleaned datasets were loaded into SQLite and used for analytical processing and dashboard generation.

Workflow:

Raw Data → Cleaning → SQLite → Analytics → Power BI Dashboard

5. Data Cleaning Process

Several data quality checks were performed.

Activities Performed

- Removed duplicate records.
- Standardized column names.
- Converted date columns into proper datetime formats.
- Handled missing values.
- Validated numerical fields.
- Verified consistency across datasets.

The cleaning process ensured reliable downstream analytics and dashboard performance.

6. Exploratory Data Analysis

EDA was conducted to understand trends and relationships in the mutual fund ecosystem.

Key exploration areas included:

- Industry AUM growth
- SIP inflow trends
- Category-wise fund movement
- Investor participation

- Portfolio concentration
- Benchmark performance

The analysis revealed strong growth in SIP participation and increasing mutual fund adoption.

7. Fund Performance Analysis

The scheme performance dataset was analyzed using risk-adjusted performance indicators.

Metrics Used

Sharpe Ratio

Measures excess return per unit of risk.

Alpha

Measures performance relative to benchmark expectations.

Beta

Measures volatility relative to the market.

Sortino Ratio

Measures downside-risk-adjusted return.

Observations

- Several schemes demonstrated strong Sharpe Ratios.
 - Beta values indicated varying market sensitivities.
 - Risk grades aligned with expected volatility levels.
 - High-performing schemes generally showed strong risk-adjusted returns.
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8. Investor Analysis

Investor transaction data was analyzed to identify participation patterns.

Analysis Performed

- Gender distribution
- Age group segmentation
- Transaction behavior
- Investment participation trends

Key Findings

- Investor participation is distributed across multiple demographic groups.
 - Transaction activity varies significantly among investors.
 - Certain demographic categories contribute a larger proportion of investment volume.
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9. Portfolio Holdings Analysis

Portfolio holdings were analyzed to understand sector allocation and concentration risk.

Metrics Evaluated

- Market Value
- Sector Distribution
- Stock Allocation Weights
- Portfolio Concentration

Findings

- Portfolio exposure is concentrated in a limited number of sectors.
 - Certain stocks represent a large proportion of total holdings.
 - Sector diversification varies significantly among portfolios.
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10. Benchmark Analysis

Benchmark indices were analyzed to evaluate market trends.

Analysis Areas

- Historical benchmark movement
- Comparative index performance
- Trend visualization

Findings

- Benchmark indices exhibited overall positive growth.
 - Market movements reflected broader investment sentiment.
 - Long-term performance trends remained favorable.
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11. Dashboard Development

A Power BI dashboard was developed to provide an interactive analytics experience.

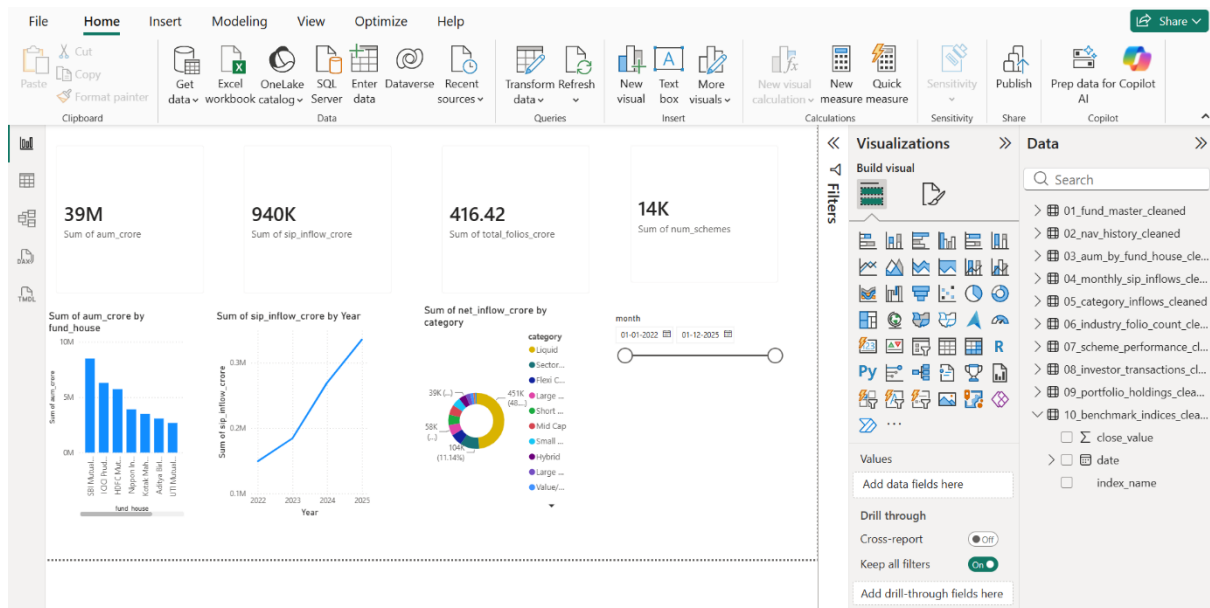
Dashboard Pages

Page 1: Industry Overview

Features:

- Total AUM
- SIP Inflows

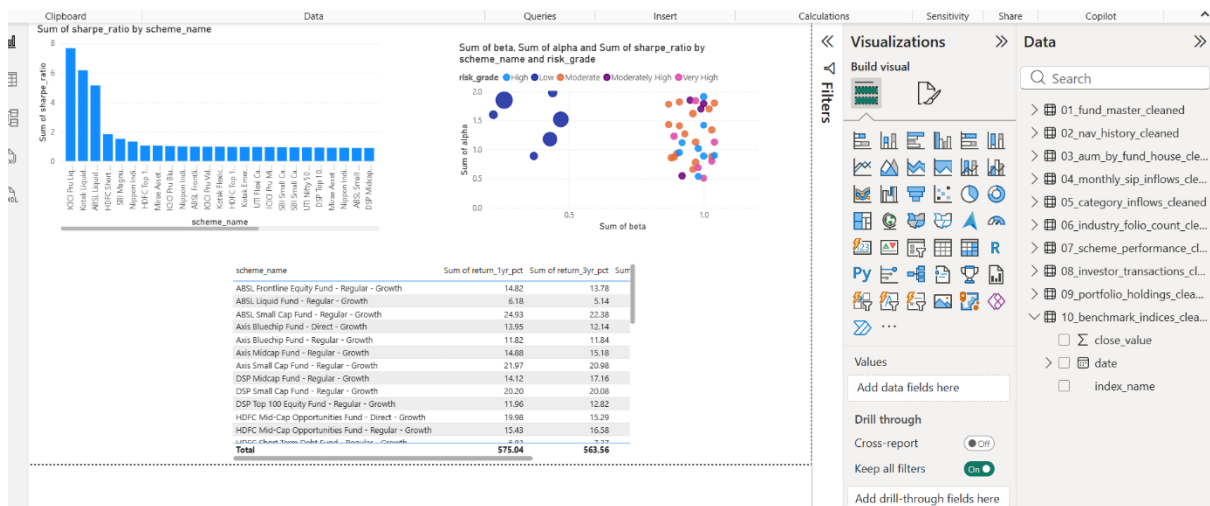
- Folio Counts
- Industry Trends



Page 2: Fund Performance

Features:

- Sharpe Ratio Analysis
- Alpha vs Beta Analysis
- Return Metrics



Page 3: Investor Analysis

Features:

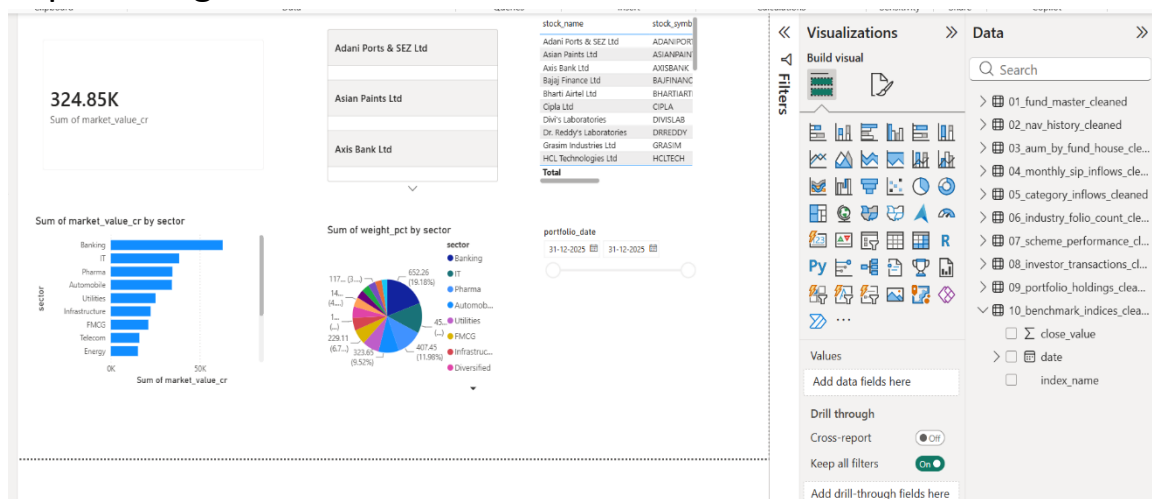
- Investor Segmentation
- Demographic Insights
- Transaction Analysis



Page 4: Portfolio Holdings

Features:

- Sector Allocation
- Market Value Distribution
- Top Holding

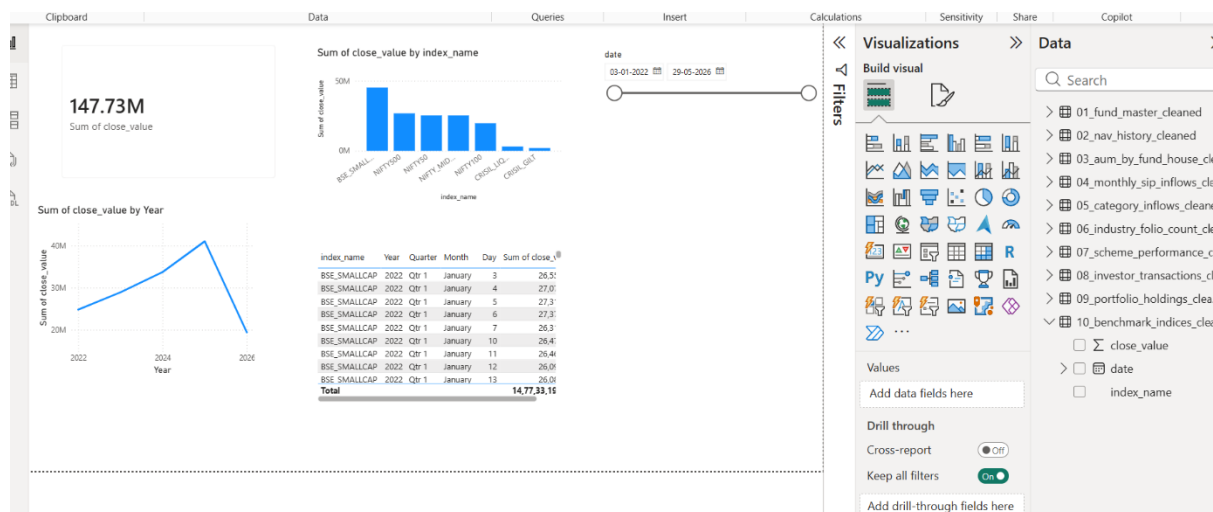


Page 5: Benchmark Analysis

Features:

- Benchmark Trend Analysis

- Index Comparisons



12. Key Findings

The project generated several important insights:

1. Industry AUM displayed sustained growth.
2. SIP inflows increased consistently across the study period.
3. Risk-adjusted performance varied significantly among schemes.
4. Portfolio allocations showed concentration in specific sectors.
5. Investor participation patterns differed across demographics.
6. Benchmark indices demonstrated positive long-term trends.

13. Limitations

The project has certain limitations:

- Limited historical depth for some datasets.
- Analysis depends on available benchmark information.
- Market conditions can change rapidly.
- Certain investor attributes may not be fully represented.

14. Recommendations

Based on the analysis:

- Increase portfolio diversification.
- Monitor risk-adjusted performance metrics regularly.
- Expand investor segmentation studies.
- Integrate real-time market feeds.
- Develop predictive analytics models for future growth forecasting.

15. Conclusion

The Mutual Fund Analytics Capstone Project successfully demonstrated the complete data analytics lifecycle, including data acquisition, cleaning, transformation, analysis, visualization, and reporting.

The project provided valuable insights into mutual fund industry trends, scheme performance, investor behavior, portfolio composition, and benchmark performance. The final Power BI dashboard offers a comprehensive and interactive platform for exploring mutual fund analytics.

This project strengthened practical skills in Python, SQL, data analysis, business intelligence, and financial analytics while delivering a professional industry-oriented solution.